Generative Adversary Modeling for Robust Convoy Defense Evaluation

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Abstract

Optimizing the ship layout of World War II convoys requires a diverse yet realistic set of U-boat torpedo attack profiles to produce meaningful results. Therefore, this project aims to train a variational autoencoder (VAE) to learn a usable latent space of attack profiles for downstream generative simulation. Additionally, advantageous U-boat attack profiles can be identified for use in tactical changes to convoy defense.

To train the VAE model, sufficient and usable synthetic training data was needed. Training data was stochastically generated under constraints based on a spawn-location-first approach with intent metadata, dynamic outcome gating, and enforced minimum ship clearances. This formed the core challenge of the project: ensuring training-manifold realism. Early centroid-based synthetic data produced tactically artificial geometries. I shifted to a spawn-first, target-aware synthetic generator (`random_tactical_v4`) with clearance constraints, then trained and sampled VAEs using latent-bank decoding.

In simulator evaluations, VAE-generated profiles were clearly stronger and more diverse than the legacy P01–P60 profile set and were competitive against direct curated synthetic sampling, with mixed 70/30 curated-random VAE candidates yielding the strongest pool-average loss metrics.

1 Introduction

The ultimate goal is to optimize the layout of WWII convoys via reinforcement learning (RL). However, convoy defense simulation requires U-boat attack profiles that are not only numerically valid but also tactically plausible. WWII convoys operated under fixed layout constraints, often with rows of seven ships and columns of six ships, as a defensive measure against U-boat torpedo attacks. By changing factors such as the number of ships in each row and column, distances between ships, and evasive maneuver patterns, a layout with a lower expected hit count may be found.

In this project, the focus is on building a generative model capable of sampling realistic U-boat attack geometries for WWII convoy scenarios. This ensures that, when optimizing the layout and dynamics of the convoy, a diverse and realistic threat profile is available for robust testing and validation.

The foundation of this project is a robust and realistic two-dimensional simulation framework that models dynamically moving convoy kinematics against active U-boats and torpedo geometries. Additionally, attack profile audits are in place to ensure U-boats spawn realistically (not on top of convoy ships, etc.) and produce plausible firing geometries (U-boats fire torpedoes toward the convoy). Attack profiles are represented with explicit fields (e.g., submarine position, bearing, spread, launch timing, speed), and each generated record can be checked for feasibility and labeled outcomes. This structure made it possible to test a data-centric hypothesis: if the synthetic generator better matches tactical reality, a simple variational autoencoder (VAE) should perform better without major architectural changes.

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Original tests were conducted with a fixed profile library (P01–P60), which worked as a baseline for testing RL code but was limited in diversity and coverage. True optimization could not be achieved with only 60 threat profiles, making a generative approach to creating additional profiles a natural extension. The goal was to learn a data manifold from synthetic but curated attack records and sample new profiles from that manifold. Here, the model class is a VAE operating on an 8-dimensional continuous feature vector (U-boat position, bearing encoding, spread, launch timing, and initial speed), with doctrine fields fixed at decode time to maintain simulator compatibility. This keeps the model simple and auditable while still enabling nontrivial candidate generation.

The main technical obstacle was creating high-quality and realistic training data rather than VAE training stability. The original synthetic data relied on centroid-biased range structures (U-boats attacked toward the center of the convoy rather than the nearest ship) and was limited to locations in which U-boats could spawn. This created visually formal and tactically narrow patterns that were unrealistic compared to historical U-boat operations. Even when training loss appeared healthy, decoded prior samples were often implausible for the intended use.

To address this, the project moved to a spawn-first, target-aware generator (`random_tactical_v4`) with explicit minimum-clearance enforcement (250 m), broader tactical regions (outside perimeter, beam, ahead, astern, and inside-column starts), and acceptance through dynamic moving-convoy outcome checks. This redesign resulted in a training distribution that was historically defensible and operationally useful for simulation.

VAEs are a standard latent-variable approach for learning compressed generative structures in continuous spaces (1). In safety-critical fields, including self-driving cars, a similar pipeline is used: generate plausible scenarios from data, enforce physical constraints, and evaluate outcomes in simulation before deployment (2). Similarly, generative adversarial networks (GANs) are often used to produce sharper scenarios and may serve as a future extension of this project to identify highly effective attack profiles.

This project follows the self-driving-car workflow but applies it to WWII convoy defense. A synthetic but doctrine-constrained generator defines the realism envelope, after which a VAE learns that manifold and proposes new attack profiles for adversarial testing. Related historical analyses of convoys focus on static factors such as the number of ships, detection range, escort effects, or air-cover benefits (4). These works provide valuable physical and strategic intuition but are largely descriptive and static because they only analyze historical data.

My previous simulation-based research into convoy size indicates that larger convoys are less likely to be detected and therefore have lower sink rates. However, it does not model defensive optimization beyond increasing convoy size and assumes U-boats search for convoys randomly. No existing work models a distribution over attacker behaviors and therefore cannot capture how adversaries might adapt or exploit weaknesses in convoy layouts.

2 Methodology

2.1 Problem Formulation

This work studies generative adversarial scenario creation for convoy defense simulation. The objective is to build upon a physics-based convoy simulator and learn a compact model capable of proposing diverse and feasible U-boat attack profiles that can be stress-tested under the same simulator used for evaluation.

Given a convoy state/layout and simulation configuration, the goal is to learn a generator G that samples attack profiles a from a realistic tactical distribution $p(a)$, subject to physical and operational constraints such as minimum spawn clearance from convoy ships. Generated attacks are evaluated by the moving-convoy simulator to estimate threat metrics including hit probability and expected loss. The central research question is whether a VAE-based generative model can produce a stronger adversarial candidate source than fixed handcrafted profile libraries and direct random sampling.

2.2 Convoy Simulation Environment

The underlying simulation framework models dynamically moving convoy kinematics against active U-boats and torpedo geometries in two dimensions. The simulator supports moving convoys, zig-zag maneuver patterns, torpedo spread geometries, and Monte Carlo evaluation across repeated stochastic trials.

Generated attack profiles are represented as structured simulator-native records containing explicit geometric and timing fields. Physical feasibility constraints are enforced during both data generation and inference. In particular, U-boats must satisfy minimum spawn-clearance requirements relative to convoy ships and must produce tactically plausible firing geometries directed toward the convoy.

Attack profiles are evaluated through repeated Monte Carlo simulation runs to estimate attacker-facing metrics such as expected hits, expected unique ships hit, expected loss, and Conditional Value at Risk (CVaR) variants.

2.3 Synthetic Attack Profile Generation

A core methodological decision was to treat training-manifold quality as a first-order concern. Early synthetic data generation relied on centroid-biased attack geometries that produced tactically rigid and unrealistic spatial patterns. These distributions lacked sufficient spawn-location diversity and resulted in decoded VAE samples that were often implausible despite low reconstruction loss.

To address this issue, the synthetic data pipeline was redesigned into the `random_tactical_v4` generator. This generator follows a spawn-first, target-aware methodology rather than a centroid-first methodology.

The revised generator incorporates the following constraints and design features:

- Multi-region tactical spawning, including outside-perimeter, beam, ahead, astern, and inside-column attack starts.
- Explicit minimum spawn-clearance enforcement of 250 m from convoy ships.
- Dynamic acceptance auditing through moving zig-zag convoy simulations.
- Tactical-outcome balancing during training-data generation:
  - 65% hit-threat profiles,
  - 25% near-miss profiles,
  - 10% miss profiles.

This redesign produced a tactically diverse and operationally defensible training distribution suitable for latent-variable learning.

2.4 VAE Architecture

The model class is a latent-variable generative model implemented as a variational autoencoder (VAE) over an 8-dimensional continuous feature vector:

$$x = \begin{pmatrix} \texttt{u\_pos\_x},\ \texttt{u\_pos\_y},\ \texttt{sin\_base\_bearing\_rad}, \\ \texttt{cos\_base\_bearing\_rad},\ \texttt{spread\_rad},\ \texttt{launch\_delay\_s}, \\ \texttt{salvo\_interval\_s},\ \texttt{u\_boat\_initial\_speed\_mps} \end{pmatrix}.$$

The encoder maps attack features into a latent distribution $q_\phi(z \mid x)$, while the decoder reconstructs attack features from latent code z. Training optimizes reconstruction loss together with KL-divergence regularization using beta-VAE-style weighting:

$$\mathcal{L} = \mathbb{E}_{q_\phi(z|x)} \left[\log p_\theta(x \mid z)\right] - \beta D_{KL} \left(q_\phi(z \mid x) \,\|\, p(z)\right).$$

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The active experimental configuration uses a compact architecture with:

- `latent_dim` $= 6$,
- `hidden_dim` $= 64$,
- $\beta = 0.03$.

The compact architecture was intentionally selected to simplify training stability analysis and latent-space diagnostics.

Several doctrine-level parameters are intentionally held fixed during decoding, including:

- moving U-boat behavior,
- four-torpedo salvos,
- uniform divergent spread,
- standard torpedo parameters.

This constraint improves simulator compatibility and allows for cleaner comparison of attack effectiveness metrics. For example, varying torpedo counts would otherwise introduce large evaluation-scale differences unrelated to attack geometry quality.

2.5 Latent-Bank Sampling Procedure

Initial experiments showed that direct Gaussian-prior sampling was insufficiently reliable for the multimodal tactical attack space. As a result, the final workflow used latent-bank sampling rather than unconstrained prior decoding.

The implemented latent-bank workflow proceeds as follows:

1. Encode real training examples into latent vectors.
2. Store empirical latent codes in a latent bank.
3. Sample near stored latent vectors using small Gaussian perturbations (typically $\sigma \approx 0.10$).
4. Decode latent samples into continuous attack features.
5. Reconstruct simulator-native attack profiles.
6. Enforce spawn-clearance and dynamic audit constraints.
7. Filter candidates by accepted tactical outcomes, typically `credible_hit_threat`.
8. Export validated candidate pools as JSONL records with diagnostics.

This process transforms the VAE into a controlled adversarial candidate proposer rather than a fully unconstrained generative sampler.

2.6 Scenario Evaluation Framework

Generation and evaluation were intentionally separated into two stages.

Generation stage: Safe and tactically plausible candidate pools are generated from the VAE model under physical and operational constraints.

Evaluation stage: Each candidate profile is evaluated through repeated Monte Carlo convoy simulations across multiple stochastic trials and seeds.

The simulator computes attacker-facing evaluation metrics including:

- `expected_hits`,
- `expected_unique_ships_hit`,
- `expected_loss`,
- CVaR-based tail-risk metrics.

2.7 Baseline Methods

All baseline methods are evaluated under the same simulator objective and scoring framework.

The primary baselines are:

- Original handcrafted P01–P60 attack-profile library,
- Direct curated synthetic sampling,
- VAE-generated candidate pools from curated, random, and mixed latent-bank sources.

The strongest observed performance was achieved through mixed 70/30 curated-random VAE candidate pools, which produced the best pool-average loss metrics during simulator evaluation.

2.8 Technical Challenges and Design Constraints

Several major technical challenges emerged during development:

- **Manifold mismatch:** low reconstruction loss can coexist with tactically implausible decoded samples.
- **Multimodality:** a single Gaussian prior is weak for highly structured tactical attack geometry.
- **Constraint satisfaction:** generated attack profiles must remain physically feasible and clearance-safe.
- **Evaluation bias risk:** poor scenario quality can distort downstream robustness conclusions.
- **Intent-field gap:** the VAE decodes continuous geometric and timing variables directly, while higher-level tactical intent metadata must be re-derived after decoding.

2.9 Methodology Flowchart

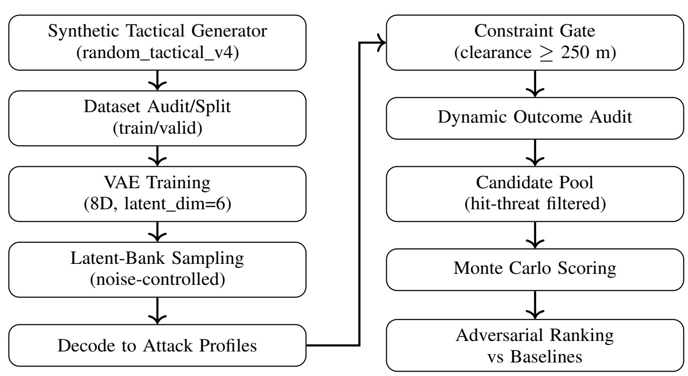

Figure 1: Adversarial scenario generation and evaluation pipeline.

3 Results

3.1 Evaluation Objective

The evaluation framework tests whether the VAE improves adversarial attack-profile generation relative to existing baselines while using the same moving-convoy simulator and scoring pipeline for all candidate sources. The primary objective is not only to maximize attack effectiveness but also to improve tactical diversity and spatial coverage relative to the original handcrafted attack library.

3.2 Experimental Configuration

The final comparison evaluates three candidate sources:

- **Original fixed library:** P01–P60 handcrafted attack profiles (60 profiles),
- **Direct curated synthetic sampling:** `random_tactical_v4` hit-threat profiles (1,000 records),
- **Mixed VAE candidate pool:** 70/30 curated-random latent-bank training mixture (1,000 hit-threat records).

All methods were evaluated under the same simulator configuration:

- convoy configuration: `convoy_layout_1`,
- 30 Monte Carlo trials per candidate,
- 3 random seeds $\times$ 10 trials per seed,
- objective preset: `balanced_default`,
- `max_hits_per_torpedo`=1,
- top-k reporting with $k = 25$.

3.3 Primary Quantitative Results

Table 1 summarizes the primary evaluation metrics across all candidate sources.

Table 1: Primary simulator evaluation results across candidate sources.

Source	**Pool Exp. Loss**	**Top-25 Exp. Loss**	**Pool Exp. Hits**	**Pool Unique Ships Hit**	**Best Candidate Loss**
P01–P60	3.5565	4.5715	2.9256	1.6200	6.7800
Direct curated v4	3.6064	6.5997	3.1739	1.6206	8.0500
Mixed 70/30 VAE	3.7561	6.6073	2.8688	1.7410	7.9817

The mixed 70/30 VAE candidate pool achieved the strongest pool-average expected-loss metric and the strongest top-25 expected-loss metric. The VAE pool also produced the highest expected number of unique ships hit, suggesting broader spatial threat coverage against convoy layouts.

Direct curated synthetic sampling produced the strongest individual candidate and the highest overall expected-hit tendency. However, the VAE-generated pool remained highly competitive while maintaining improved diversity and broader tactical coverage.

3.4 Diversity and Spatial Coverage

Diversity and coverage metrics improved substantially relative to the original fixed-profile baseline.

Table 2: Diversity and spatial coverage indicators.

Metric	**P01–P60**	**Direct Curated**	**Mixed VAE**
500 m occupied spatial bins	41	370	410
Mean clearance (m)	1049.2	1546.1	1564.9
Minimum clearance (m)	—	284.2	250.5

As shown in Figure 2, the legacy P01–P60 profiles occupy a comparatively narrow spatial manifold relative to both curated synthetic and VAE-generated candidate pools.

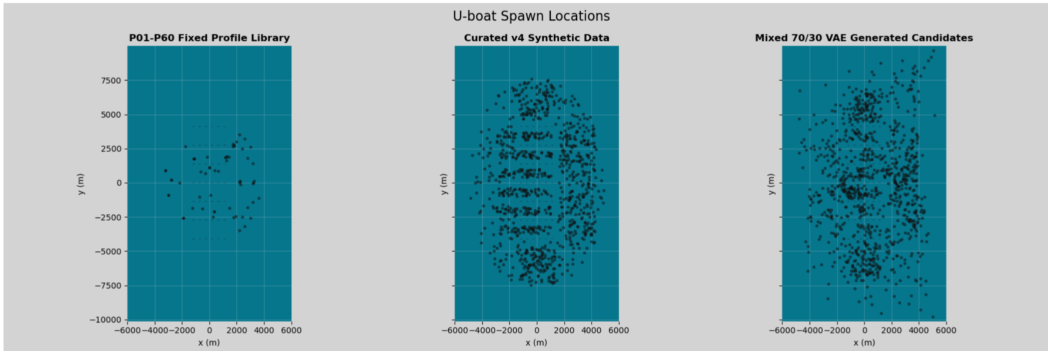

Figure 2: Spatial comparison of attack-profile candidate sources. The original P01–P60 handcrafted library occupies a narrow and highly structured spatial manifold, while both the curated synthetic generator and mixed VAE candidate pool produce substantially broader tactical coverage. The mixed VAE additionally preserves realistic clearance constraints while expanding attack-space diversity.

The mixed VAE candidate pool produced the broadest spatial coverage, occupying 410 distinct 500 m spatial bins compared to only 41 bins for the original handcrafted baseline. Both generated candidate pools maintained constraint-consistent minimum clearances above the enforced 250 m threshold.

These results indicate that the VAE successfully learned a broader tactical attack manifold than the fixed-profile library while preserving physical feasibility constraints.

3.5 Training Stability

VAE training remained stable across all experimental runs. All 45k/5k train-validation splits were trained for 60 epochs, with final validation losses converging within a narrow range of approximately 0.322–0.332.

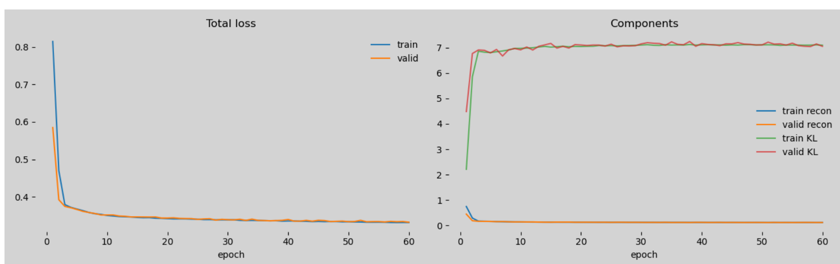

Figure 3: Training and validation loss curves for the mixed 70/30 VAE configuration across 60 epochs. Loss convergence remained stable across runs, indicating that downstream performance differences were primarily driven by candidate quality and distributional coverage rather than optimization instability.

3.6 Interpretation of Results

Three primary conclusions are supported by the evaluation results.

First, both curated synthetic sampling and VAE-generated candidate pools substantially outperform the legacy P01–P60 handcrafted profile library in terms of adversarial effectiveness and tactical diversity. The original fixed-profile baseline exhibited comparatively weak spatial coverage and lower expected-loss performance.

Second, the mixed 70/30 VAE candidate pool proved competitive with direct curated synthetic sampling and achieved slightly stronger pool-average expected-loss performance, stronger top-k loss metrics, and higher unique-ship impact.

Third, direct curated synthetic sampling still produced the strongest individual attack candidate and the highest raw hit-rate tendency. Consequently, the most defensible interpretation is not that the

VAE replaces realism-constrained synthetic generation, but rather that it complements it. The VAE performs best as a compact generative candidate source operating on top of a realism-constrained simulator and data-generation pipeline.

4 Conclusion

The results indicate that a VAE model can serve as an effective generator for creating adversarial attack profiles for use in convoy-defense optimization. However, generation must remain constrained by physical feasibility and validated through dynamic simulator outcomes in order to satisfy historical-realism requirements.

During experimentation, it became clear that training-manifold realism mattered more than increasing model complexity. Early centroid-biased synthetic data produced tactically rigid attack geometries and limited spatial diversity. Once a spawn-first, target-aware generation methodology was adopted, candidate quality improved substantially relative to the fixed P01–P60 baseline and remained competitive with direct curated synthetic sampling. These findings suggest that the VAE improves attack-space coverage and adversarial diversity, but does not replace the need for realism-constrained doctrine generation.

Future work will connect the candidate generator to a partially observable Markov decision process (POMDP) attacker framework. This extension is intended to improve realism by forcing U-boats to operate under noisy observations and belief-state decision-making rather than full-state ranking access. Under this formulation, attacker decisions would reflect incomplete information and uncertainty regarding convoy position and state, more closely matching historical operational conditions faced by WWII U-boats.

With the VAE now functioning as a curated and diverse attack-profile generator, the project can return to the original long-term objective of convoy-layout optimization through reinforcement learning. The revised workflow enables defender optimization against a more realistic and distributional attacker model: optimize convoy layout and escort policy against a constrained generative attack process (VAE + POMDP attacker), rather than against static or randomly generated attack profiles alone.

A Code Availability

The convoy simulation framework, synthetic data generators, VAE implementation, evaluation pipeline, and supporting notebooks used throughout this work can be found below.

Main repository:

https://github.com/MAP9900/Convoy-Layout-Project

VAE documentation and implementation notes:

https://github.com/MAP9900/Convoy-Layout-Project/blob/main/docs/VAE.md

All notebooks used:

https://github.com/MAP9900/Convoy-Layout-Project/tree/main/notebooks

B Additional VAE Distribution Diagnostics

The following figures compare training-data distributions against VAE-generated candidate distributions across the curated, random, and mixed latent-bank configurations. These supplementary diagnostics provide additional qualitative evidence that the VAE preserves broad spatial structure and tactical diversity while remaining within the realism-constrained manifold established by the synthetic generator pipeline.

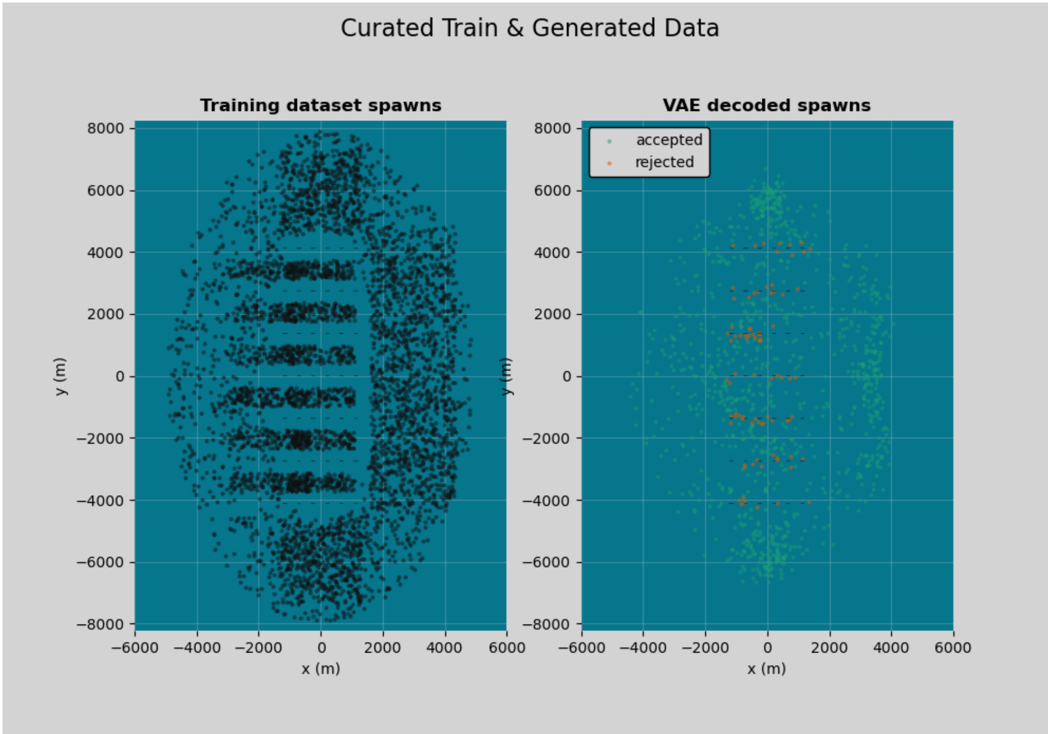

Figure 4: Comparison between curated synthetic training data and VAE-generated candidate distributions for the curated latent-bank configuration.

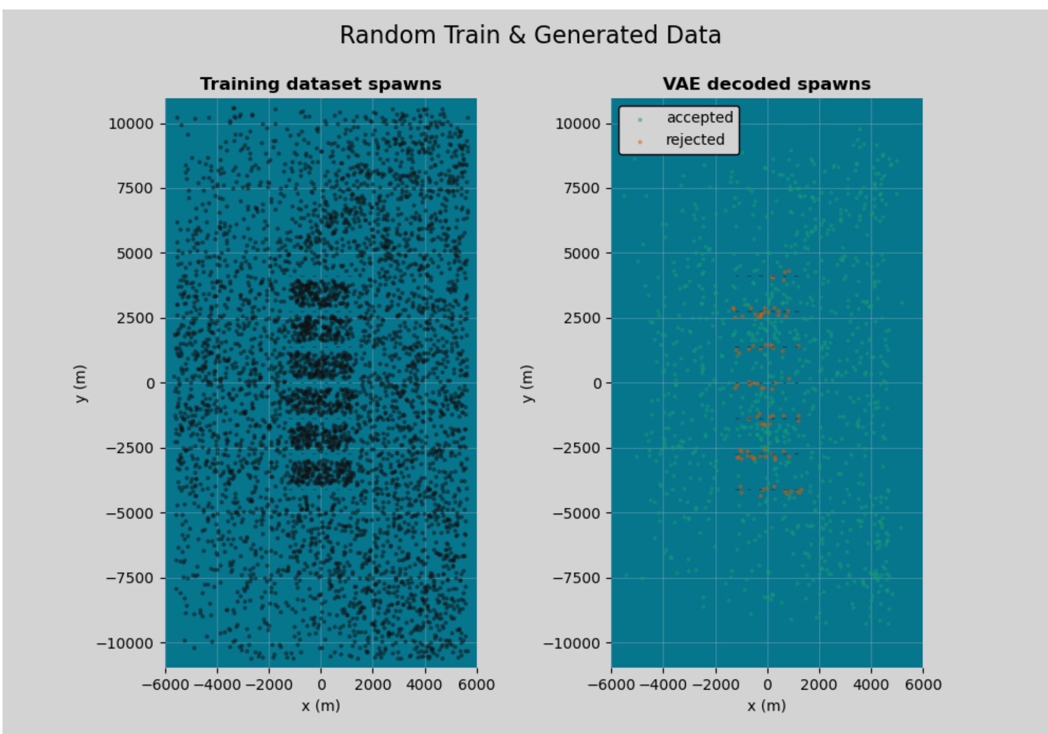

Figure 5: Comparison between random synthetic training data and VAE-generated candidate distributions for the random latent-bank configuration.

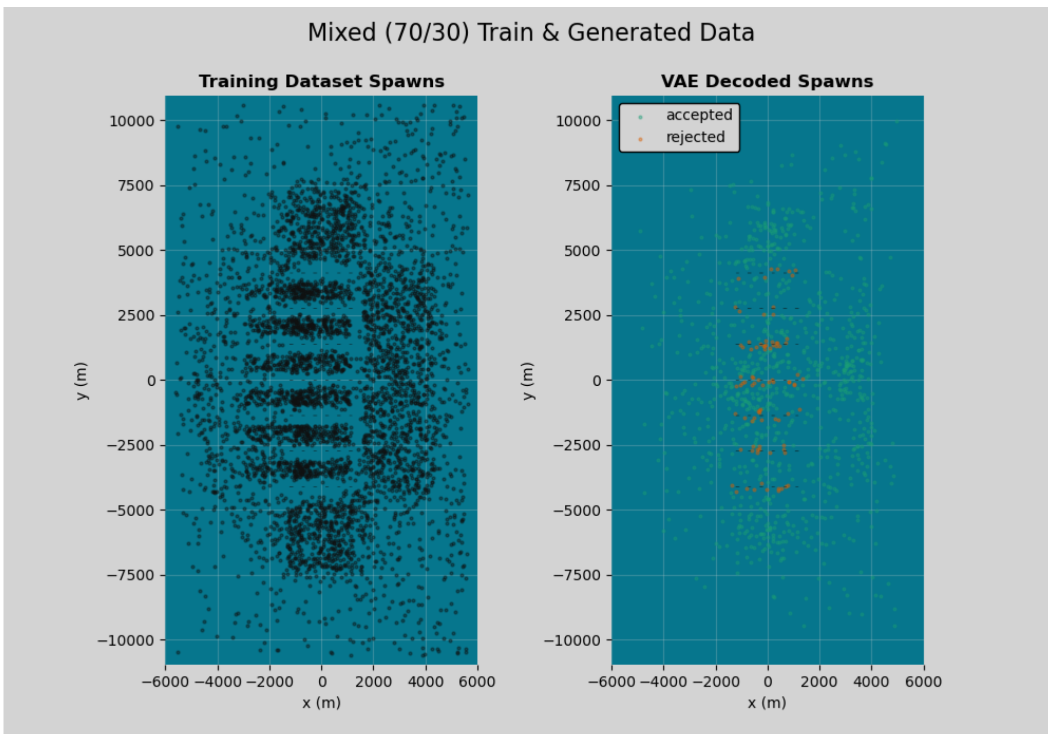

Figure 6: Comparison between mixed 70/30 training data and VAE-generated candidate distributions for the mixed latent-bank configuration.

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